GigShield

AI-Powered Parametric Insurance for India's Gig Workers

Like the traditional Tamil சீட்டு —
community-powered, automated & instant.

Food Delivery

Zomato · Swiggy

TEAM

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Guidewire DEVTrails 2026

Phase 1 Submission

The Problem

India's gig workers have no income protection against external disruptions

20–30%

monthly income lost
during disruptions

Source: Guidewire DEVTrails 2026

No Weather Protection

Rain stops deliveries. Zero income, zero safety net.

No Curfew Coverage

Unplanned lockdowns mean instant earnings loss.

No Savings Buffer

Workers live week-to-week with no financial cushion.

No Insurance Access

Traditional insurers completely ignore gig workers.

Real-Life Impact

- A delivery partner loses ₹500–₹1000 per day during disruptions
- No work = no pay, immediate income loss

15 Million+ gig workers in India — none have income disruption insurance

Sources: [1] Guidewire DEVTrails 2026 · [2] NITI Aayog Gig Economy Report, June 2022, Pg.25 Table 15 · [3] ILO India Gig Worker Report 2022

Why Current Systems Fail

Existing solutions don't work for gig workers

No Income Protection

Insurance covers health/accident/vehicle — not lost wages.

Built for Salaried Workers

Existing policies assume fixed monthly income.

No Real-Time Data

Systems don't use weather, location, or live event feeds.

Manual Claims Process

Workers apply manually. Takes days or weeks to approve.

Too Complex to Use

Confusing documents — most workers give up and don't claim.

No Instant Payout

Money arrives too late. The damage is already done.

"Current systems react late. GigShield acts instantly — before the worker even asks."

Market Opportunity

Massive, underserved, and growing fast

15M+

Gig workers in India

As of FY 2024–25

23M

Expected by 2030

55% growth in 4 years

₹0

Income protection today

Zero existing solutions

Phase 1 Target Market

Food Delivery Workers

Zomato & Swiggy partners

Tier 1 & 2 Cities

Chennai, Bangalore, Mumbai

High Rain Exposure Zones

Most disruption-affected

Android Phone Users

Web app — no install needed

"A small weekly premium from millions creates a large, sustainable protection pool."

Real-World Scenario

Who we protect and how it works in practice

Our Persona

Food Delivery Partner

Zomato / Swiggy

- Earns ₹2,500–5,000/week
- Works 8–12 hrs/day outdoors
- Highly rain-sensitive work
- No formal employment benefits
- Paid weekly by the platform
- Uses Android phone for orders

Sample Scenario — Rajan from Chennai

MON	Earns ₹700. GigShield auto-deducts ₹60 → community pool.
TUE	Heavy rain (82mm) detected in Chennai — IMD + OpenMeteo.
TUE	AI confirms disruption (94% confidence). Fraud check: PASSED.
TUE	₹300 auto-sent to Rajan's UPI. He did absolutely nothing.
SUN	Contributed ₹60 — received ₹300 back. Net gain: ₹240.
NET	Without GigShield: Rajan loses ₹300 with zero recourse.

Application Workflow

6 steps — worker does nothing after Step 1

01 Worker Onboarding

Register via web app.
Name, phone, district, platform, earnings.

→

02 AI Risk Profiling

District risk score from
10+ years of weather data.

→

03 Policy Creation

₹20–60/week deducted from
platform payout every Monday.

04 Real-Time Monitoring

Weather + news + curfew APIs
checked every 30 minutes.

→

05 Auto Claim Trigger

Disruption detected?
All zone workers identified.

→

06 Instant UPI Payout

Fraud check passes →
Payout sent in under 30 seconds.

Platform: WEB APP — No install needed. Works on any Android browser. No bank details required.

System Architecture

Clean pipeline from data input to payout output

1

External Data Sources

OpenMeteo · IMD · GDELT News · Location APIs

2

Data Processing Layer

Cleaning · Filtering · Feature Extraction

3

AI Risk Model

Risk Score (0–130) · ML on 10+ years of weather data

4

Decision Engine

Multi-Source Validation · Confidence threshold ≥ 80%

5

Auto Claim Trigger

Zone-Based Worker Identification · Batch Processing

6

Fraud Detection Engine

GPS Check · Duplicate Detection · Activity Gap

7

Instant UPI Payout

Auto-transfer to worker's UPI within 30 seconds

Weekly Premium Model — சீடு Style

Community-pooled, dynamic, and fair

Premium = Weekly Earnings × Risk Rate × Platform Multiplier × Season Multiplier

High Risk (≥100)	5% → ₹60/wk
Medium Risk (≥70)	3.5% → ₹40/wk
Low Risk (<70)	2% → ₹20/wk
Based on 10+ yrs data	

Zomato / Swiggy	1.1×
Zepto / Blinkit	1.0×
Amazon / Flipkart	0.9×
Dunzo	1.05×

Monsoon (Jun–Sep)	1.5×
Summer (Mar–May)	1.2×
Autumn (Oct–Nov)	0.9×
Winter (Dec–Feb)	0.8×

சீடு Pool Model: All premiums go into one community pool. Payouts come from the pool — not company funds. Surplus rolls to next week. Workers never lose their contribution.

Parametric Triggers — Zero-Touch Auto-Claim

Disruption detected = payout sent. Worker does nothing.

Heavy Rainfall

Trigger **> 50mm / 24h**

Source OpenMeteo API

HIGH **Payout: 75% daily**

Extreme Heat

Trigger **> 42°C**

Source IMD API

MEDIUM **Payout: 50% daily**

High Wind Speed

Trigger **> 35 km/h**

Source OpenMeteo API

MEDIUM **Payout: 50% daily**

Flood Warning

Trigger **Official IMD alert**

Source IMD Alerts API

HIGH **Payout: 75% daily**

Govt Curfew

Trigger **Movement restricted**

Source GDELT News API

HIGH **Payout: 75% daily**

Transport Strike

Trigger **Delivery zones blocked**

Source Local Auth API

MEDIUM **Payout: 50% daily**

All 6 triggers are fully automated — workers receive payouts without filing any claim

AI / ML Integration Plan

Four intelligent systems working together

01 Dynamic Risk Profiling

- 10+ years of IMD weather data analyzed
- District risk score: 0 to 130
- Weights: Rainfall 40% · Max Rain 25% · Wind 20% · Disruption Rate 15%

02 Premium Calculation ML

- Real-time season & platform multipliers
- Weekly auto-recalibration per district
- Premium capped: Min ₹20 / Max ₹60 per week

03 Fraud Detection Engine

- Duplicate claim check — 24h window
- GPS location vs registered zone validation
- Claim velocity + platform activity gap analysis

04 Parametric Auto-Trigger

- Multi-API monitoring every 30 minutes
- 2+ sources must confirm disruption
- Confidence ≥ 80% required before payout

Dataset: Kaggle — Indian Rainfall & Weather Data · Columns: date_of_record, district, state, rainfall, avg_temp, max_temp, wind_speed

Market Crash — Adversarial Defense & Anti-Spoofing Strategy

Scenario: 500 delivery partners used fake GPS to file simultaneous claims — draining the community pool. How does GigShield stop this?

Detect the Faker

- GPS trail continuity — location must match physical movement history
- Accelerometer vs GPS — phone sensor must match claimed outdoor motion
- Platform activity gap — worker must have been online before disruption

Catch the Ring

- Timing cluster — 15%+ of a zone filing within 10 min = batch hold
- Referral network graph — fraud rings share registration metadata
- Device fingerprint — same IMEI with different SIM cards flagged

Protect the Honest

- Tier 1 (trusted workers) — auto-approved instantly, no delay
- Tier 2 (new workers) — 2-hr soft hold with one-tap confirmation
- Tier 3 (flagged) — manual review within 24h, worker notified

Key Insight: Is the disruption real? (Weather API) · Was THIS worker actually affected? (Behavioral Data) — Both must be YES.

Historical Data Analysis — District Risk Profiling

Real data from Kaggle powering our premium model

Source

Kaggle — Indian Rainfall
& Weather Prediction Data

Format

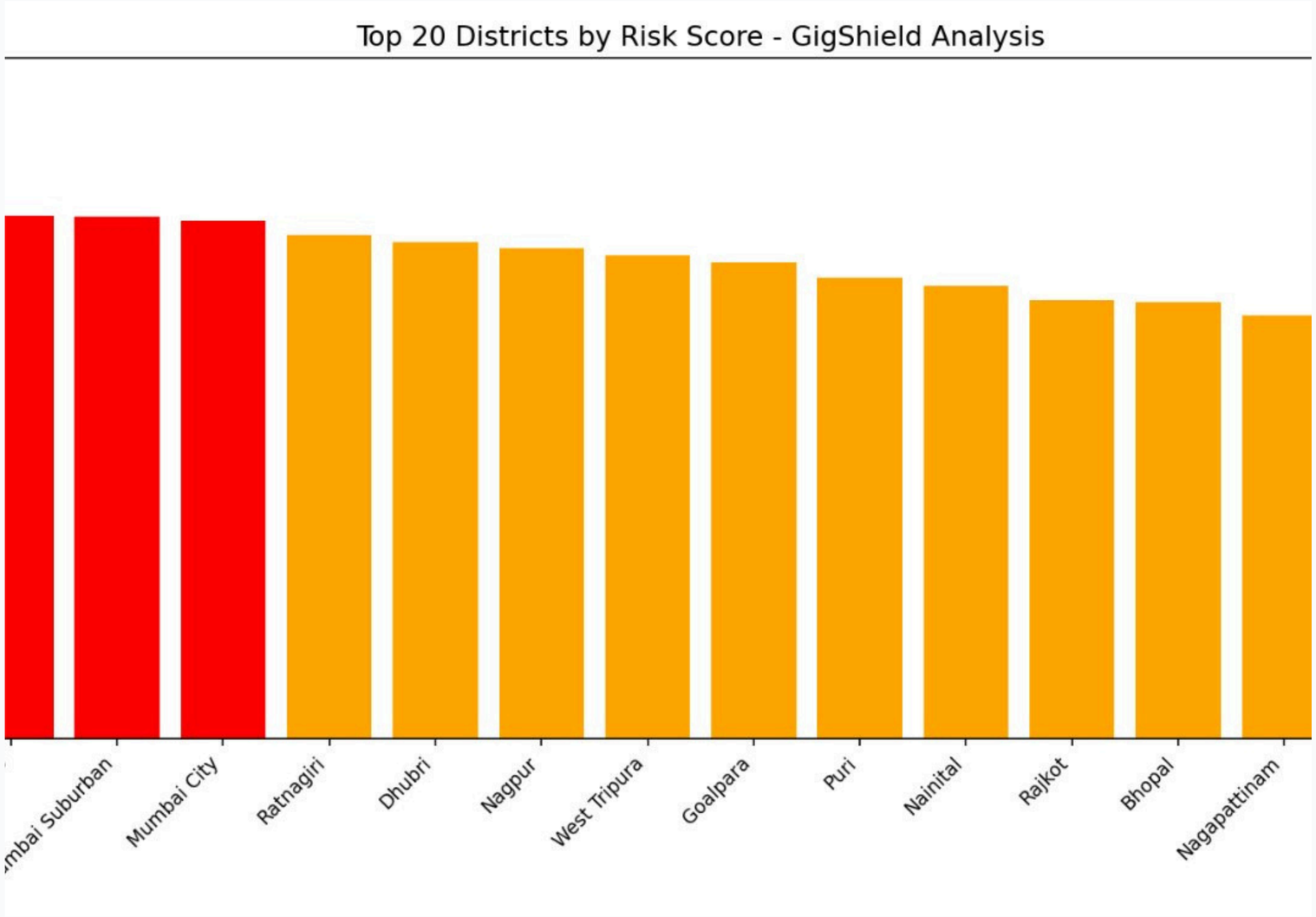
XLSX → pandas read_excel()

Columns

district, rainfall,
avg_temp, wind_speed

Link

kaggle.com/ameydilipmorye/
indian-rainfall-and-weather-data



Tech Stack & Development Plan

Simple, fast, and built for low-end devices

- Python 3.11
- Flask REST API
- Pandas + NumPy
- Scikit-learn (ML)
- SQLite / PostgreSQL

- HTML5 + CSS3
- Vanilla JavaScript
- Chart.js
- Mobile-first design
- No framework — fast load

- OpenMeteo (weather)
- IMD India (alerts)
- GDELT (news)
- Razorpay mock UPI
- Platform APIs (mock)

- Kaggle weather dataset
- district_risk_profile.csv
- Risk scoring model
- Fraud detection rules
- Pandas data pipeline

GigShield

Protecting India's Delivery Heroes

Like சீட்டு — community-powered, automated, and instant.

GitHub: https://github.com/Dhanapriya11/guidewire_hack

₹20–60
weekly premium

6
auto triggers

< 30s
payout time

Thank You!

Guidewire DEVTrails 2026 · Phase 1